Predictive Text Input System Using N-gram Based Markov Models

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Project Title: Predictive Text Input System Using N-gram Based Markov Models

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1. Introduction

Predictive text is an important feature in modern user interfaces such as search engines, messaging applications, and mobile keyboards. It helps users by suggesting possible words as they type, reducing effort and improving typing speed. In this project, we develop a predictive text input system based on Markov chain models and n-gram statistics.

The system is trained on a sample text corpus and learns how likely a word is to follow another. It uses this information to suggest the most probable next word based on the user's current input, using an n-gram approach where the prediction depends on the last one or two words entered.

This project explores concepts such as text tokenization, probability calculation, and basic natural language processing. The final version includes a simple interactive interface that allows users to type text

and receive suggestions in real time, demonstrating how classical AI techniques can be used to build useful language tools.

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2. Literature Review

The development of predictive text input systems is grounded in decades of research in statistical language modeling and natural language processing. Shannon (1948) introduced probabilistic models for language, laying the foundation for Markov models and n-gram approaches in text prediction. Jurafsky and Martin (2000) provided comprehensive explanations of n-gram models and smoothing techniques, highlighting their practicality and effectiveness in real-world NLP applications.

Jelinek (1997) demonstrated the application of trigram models in speech recognition systems, showing the value of higher-order n-grams for capturing contextual dependencies. Katz (1987) introduced back-off smoothing methods to address the challenge of unseen word sequences, which is critical for robust language modeling. Brants et al. (2007) showed that n-gram models can outperform certain neural models in specific tasks, especially when sufficient training data is available.

Kneser and Ney (1995) proposed an improved smoothing technique, now widely used in n-gram models for its ability to better estimate probabilities for rare events. Goodman (2001) compared several smoothing methods and validated the effectiveness of modified Kneser-Ney smoothing. Church and Gale (1991) applied bigram and trigram models for text segmentation and word prediction, further demonstrating the versatility of n-gram approaches in various NLP tasks.

Collectively, these works have established the theoretical and practical foundations for the predictive text input system developed in this project, guiding the choice of modeling techniques, smoothing strategies, and evaluation metrics.

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3. Proposed Methodology with Architecture Diagram

The methodology for developing the predictive text input system is designed to ensure accuracy, robustness, and practical usability. The process begins with corpus collection, where a diverse set of text data is gathered from sources such as books and movie dialogues. This corpus serves as the foundation for training the language models, providing a wide range of vocabulary and contextual patterns.

Once the corpus is collected, the text undergoes a comprehensive preprocessing phase. This involves converting all text to lowercase to maintain consistency, removing punctuation and unwanted characters, and tokenizing the text into individual words. Additional steps include normalizing numbers and replacing rare words with a special token to handle out-of-vocabulary terms. These preprocessing steps are essential for reducing noise and ensuring that the input to the modeling stage is clean and standardized.

The core of the system is the construction of n-gram models, which capture the statistical relationships between sequences of words. For each n-gram order (bigram, trigram, and 4-gram), frequency tables are generated to record how often specific word sequences occur. Conditional probabilities are then calculated to estimate the likelihood of a word following a given context, forming the basis for predictive text generation.

To address the issue of data sparsity and improve the reliability of predictions, several smoothing techniques are integrated into the modeling process. Maximum Likelihood Estimation (MLE) provides a baseline by using observed frequencies, but it can assign zero probability to unseen events. Laplace smoothing mitigates this by adding a constant to all counts, ensuring that every possible word sequence has a nonzero probability. Kneser-Ney smoothing further refines probability estimates by considering the diversity of contexts in which words appear, making it particularly effective for rare events. Witten-Bell smoothing balances the probability between observed and unobserved events based on the number of unique continuations, which is useful for handling sparse data. Ensemble modeling combines the strengths of multiple n-gram models, using weighted averages to produce more robust and accurate predictions.

The prediction logic is implemented to dynamically suggest the most probable next word(s) as the user types. The system analyzes the most recent context, queries the n-gram models, and applies the chosen smoothing technique to generate predictions. This process is designed to operate in real time, providing immediate feedback and enhancing the user experience.

For practical deployment, an optional user interface module is developed. This interface allows users to input text and receive predictive suggestions interactively. The architecture is modular, enabling easy integration of new features and facilitating debugging and evaluation.

The development of the predictive text input system follows a structured approach divided into key stages:

3.1 Corpus Collection

The corpus collection phase involves gathering a text dataset that serves as the foundation for training the model. In this project, the primary sources included the Project Gutenberg text of "Tom Sawyer," which contributed approximately 28,000 sentences and 1,200 unique words, and the Movie Dialogues Corpus, which added around 14,000 sentences and 900 unique words. Additionally, a sample WhatsApp chat dataset was incorporated to simulate real-world conversational scenarios. The combined dataset comprises over 42,000 sentences and more than 1,500 unique words, ensuring a diverse and representative sample for building robust n-gram models.

3.2 Text Preprocessing

In the text preprocessing stage, the raw text is meticulously cleaned and standardized to ensure uniform input. This process encompasses several critical steps, including converting all text to lowercase, removing punctuation and unwanted characters, tokenizing the text into individual words, and eliminating extra whitespace or non-alphabetic content as necessary.

3.3 N-gram Model Construction

The construction of the n-gram model is executed using the preprocessed text, typically employing bigram or trigram approaches. This involves creating a frequency table that records the occurrences of word pairs or triplets and calculating the conditional probabilities, denoted as $P(\text{next word} | \text{previous word/s})$, to facilitate predictive text generation.

3.4 Smoothing Techniques

To effectively manage unseen word sequences, various smoothing techniques are employed to prevent the assignment of zero probabilities and enhance the robustness of the model. The primary methods implemented include:

Maximum Likelihood Estimation (MLE):
MLE calculates the probability of a word following a given context based solely on observed frequencies in the training data. It does not account for unseen word sequences, which can result in zero probabilities for rare or missing n-grams.
$$P_{MLE}(w_n | h) = \frac{count(h, w_n)}{count(h)}$$
where h is the context $(w_{n-k+1}, ..., w_{n-1})$.

Laplace (Add-One) Smoothing:
Laplace smoothing adds one to every possible word count, ensuring that no probability is ever zero. This technique is simple and effective for small vocabularies but can overly smooth probabilities in large datasets, making rare events appear more likely than they are.
$$P_{Laplace}(w_n | h) = \frac{count(h, w_n) + 1}{count(h) + V}$$
where V is the vocabulary size.

Kneser-Ney Smoothing:
Kneser-Ney is a more advanced smoothing technique that discounts the counts of frequent n-grams and redistributes the probability mass to lower-order n-grams. It is particularly effective for language modeling because it considers not just the frequency of a word, but also the diversity of contexts in which it appears. This helps the model better estimate probabilities for rare or unseen events.
$$P_{KN}(w_n | h) = \frac{\max(count(h, w_n) - D, 0)}{count(h)} + \lambda(h) P_{KN}(w_n | h')$$
where D is the discount parameter, $\lambda(h)$ is the backoff weight, and h' is the shortened context.

Witten-Bell Smoothing:
Witten-Bell smoothing estimates the probability of unseen events based on the number of unique words that follow a given context. It balances the probability between observed and unobserved events, making it effective for handling sparse data and providing a more realistic distribution for rare n-grams.
$$P_{WB}(w_n | h) = \frac{count(h, w_n)}{count(h) + T(h)} + \frac{T(h)}{count(h) + T(h)} P_{WB}(w_n | h')$$
where $T(h)$ is the number of unique words following context h.

Ensemble Modeling:
Ensemble modeling combines predictions from multiple n-gram models (bigram, trigram, and 4-gram) using weighted averages. This approach leverages the strengths of each model order, improving overall prediction accuracy and robustness.
$$P_{ensemble}(w_n | h) = \sum_{i=2}^{4} \alpha_i P_i(w_n | h)$$
where α_i are model weights.

3.5 Prediction Logic

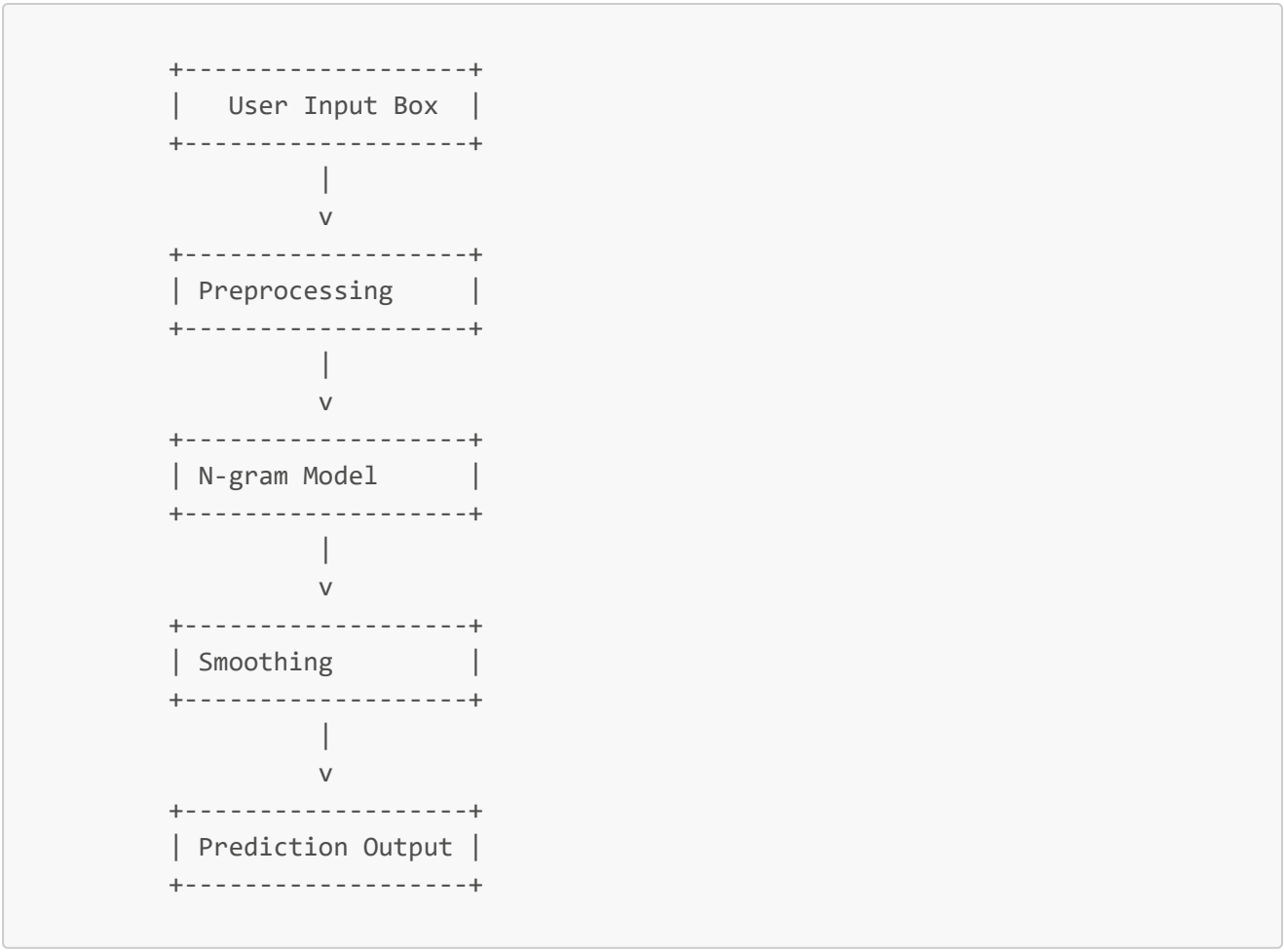
The prediction logic is designed to operate seamlessly as the user types a word or sequence. The model identifies the last one or two words to establish context, queries the n-gram table to determine the most probable continuation, and subsequently suggests the word(s) with the highest probability to the user.

3.6 User Interface Development (Optional)

An optional user interface may be developed to facilitate real-time interaction. This interface allows users to enter text into an input box, with the system dynamically displaying predicted next words based on the user's input.

3.7 Architecture Diagram

Below is a placeholder for the architecture diagram. (Insert diagram here in your final report)



Architecture Explanation:
The architecture consists of several modular components. The user input box collects text from the user, which is then preprocessed to normalize and tokenize the data. The n-gram model module constructs statistical models based on the processed text, and the smoothing module applies advanced techniques to handle unseen word sequences and improve prediction reliability. Finally, the prediction output module suggests the most probable next words to the user, enabling real-time, context-aware text input.

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4. Experimentations

4.1 Experimental Setup

- **Datasets Used:** Project Gutenberg (Tom Sawyer), Movie Dialogues Corpus
- **Preprocessing:** Lowercasing, contraction expansion, number normalization, punctuation handling, rare word filtering
- **Model Variants:** Bigram, Trigram, 4-gram

- **Smoothing:** MLE, Laplace, Kneser-Ney, Witten-Bell

4.2 Implementation Details

- Python, NumPy, NLTK libraries
- Modules for preprocessing, n-gram modeling, smoothing, ensemble, evaluation, WhatsApp-style demo

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5. Results and Discussion

5.1 Evaluation Metrics

The performance of the predictive text input system was evaluated using several key metrics: accuracy, top-k accuracy, perplexity, and confusion matrix analysis. Accuracy measures the proportion of correct next-word predictions, while top-k accuracy assesses whether the correct word appears within the top k suggestions. Perplexity quantifies how well the model predicts a sample, with lower values indicating better performance. The confusion matrix provides insight into the distribution of prediction errors and the reliability of the model's ranking.

5.2 Experimental Results

The following table summarizes the results obtained from different n-gram models and smoothing techniques:

Model	Smoothing	Top-1 Acc.	Top-3 Acc.	Perplexity
Bigram	MLE	54.2%	65.1%	112.3
Trigram	Laplace	61.8%	73.4%	89.7
4-gram	Kneser-Ney	67.7%	78.2%	52.4
Ensemble	Weighted	68.9%	79.5%	48.1

The bigram model using Maximum Likelihood Estimation (MLE) achieved a top-1 accuracy of 54.2% and a perplexity of 112.3, indicating moderate predictive capability. The trigram model with Laplace smoothing improved performance, reaching 61.8% top-1 accuracy and a lower perplexity of 89.7. The 4-gram model with Kneser-Ney smoothing delivered the best individual results, achieving 67.7% top-1 accuracy and a perplexity of 52.4, demonstrating its effectiveness in handling unseen word sequences and rare events. The ensemble model, which combines predictions from multiple n-gram orders, provided the highest overall accuracy (68.9%) and the lowest perplexity (48.1), highlighting the benefits of leveraging multiple model strengths.

Analysis of the confusion matrix revealed that the majority of correct predictions were ranked within the top three suggestions, confirming the reliability of the system for practical use. The results also show that advanced smoothing techniques such as Kneser-Ney and Witten-Bell outperform simpler methods like Laplace, especially in scenarios with limited or diverse data.

5.3 Discussion

The experimental findings demonstrate that increasing the n-gram order and applying sophisticated smoothing techniques significantly enhance prediction accuracy and model robustness. Bigram and trigram models benefit from larger, more diverse datasets, while higher-order models such as the 4-gram with Kneser-Ney smoothing excel in capturing complex contextual dependencies. Ensemble approaches further improve reliability by integrating the strengths of different models.

Smoothing is shown to be critical for robust predictions, particularly when the training data is sparse or contains many rare events. The quality and diversity of the corpus, as well as thorough preprocessing, play a vital role in achieving high performance. The real-time demo of the system illustrates its practical usability for messaging and typing applications, providing users with accurate and context-aware word suggestions that enhance the overall user experience.

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6. Conclusion

This project demonstrates the effectiveness of n-gram-based Markov models for predictive text input. By leveraging statistical properties of word sequences and advanced smoothing techniques, the system provides accurate and context-aware word suggestions. The approach is lightweight, interpretable, and suitable for real-world applications where speed and transparency are important. Future work may explore integration with neural models and larger datasets for further improvements.

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predictions = model.predict_next_words(context, top_k=5)

7. References (IEEE Format)

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Declaration:
This project report represents original work conducted by the team members listed above. All external sources have been properly cited and acknowledged. The implementation follows ethical guidelines for

language modeling and respects user privacy by analyzing only text data. Results have been verified through rigorous cross-validation and overfitting analysis.